Spatial distance and curvature from an exact counting identity in two-dimensional causal sets

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In causal set theory, geometry must be recovered from the causal order and from counting. In two dimensions, the causal-overlap regions of Boguñá and Krioukov satisfy an exact identity: for two spacelike elements and *any* element c in their common past, $d^2 = 2V_AV_B/V_C$ in flat spacetime. With a Poisson bias correction it gives a spatial-distance estimator that is unbiased for d^2 above two discreteness lengths and, on the same pairs, more accurate than the equal-depth and past–future constructions below about ten discreteness lengths; at larger separations the past–future construction of Hu, Meyer and Meyer is more accurate. The freedom in the choice of c, not the bias correction, is what improves on the equal-depth estimator. In curved spacetime the identity fails in a way that depends on the depth of c. In de Sitter space $d^2_{\rm est}/d^2 = \tanh^2(H\tau/2)/[2\ln\cosh(H\tau/2)] = 1 - R\tau^2/16 + O(\tau^4)$, and in any $1+1$ FRW spacetime the leading failure measures the scalar curvature one third of the way from the pair to c. Pooled over pairs, a fit of this depth drift with a window defined from counts alone is an unbiased curvature estimator on sprinklings with inserted pairs. We compare its noise with the Benincasa–Dowker operator and with chain abundances at equal density. As a supporting diagnostic we calibrate the Lorentzian-embedding test of Clough and Evans and add the fraction of null-separated pairs, which lets the test reject layered orders on its own. A Laguerre identity for the mean of Johnston's propagator at fixed interval size is given in an appendix.

I. INTRODUCTION

A causal set is a locally finite partially ordered set: every interval $I(x, y) = \{z : x \prec z \prec y\}$ is finite [1, 2]. Causal set theory proposes that spacetime is such a set, with the continuum emerging at large scales. The number of elements in a region then measures its spacetime volume and the order fixes the light cones; the kinematic programme is to recover all other geometric information from these two ingredients.

Part of this programme is well established. The longest chain between two elements estimates their proper time [3]; interval and chain abundances give the dimension [4, 5] and the curvature [6, 7]; the Benincasa–Dowker operator gives the scalar curvature and the action [8, 9]. Spatial distance is harder, because spacelike elements bound no interval. Boguñá and Krioukov measured it through the overlap of the pasts of two elements as seen from an element c at equal proper time below both [10]; Brightwell and Gregory and Rideout and Wallden used the smallest interval containing both elements [3, 11]; Hu, Meyer and Meyer gave an exact reconstruction in $1 + 1$ dimensions from a past and a future element [19]. Whether a causal set resembles a manifold at all is tested by dimension estimators, interval abundances [7] and Lorentzian embeddings [20, 22, 23]. Matter can be introduced through Johnston's sum over chains [24, 25], whose average over sprinklings reproduces the Klein–Gordon propagator.

Two-dimensional causal sets are a natural laboratory: every $1+1$ metric is conformally flat, intervals are rectangles in light-cone coordinates, and a sprinkled diamond is a uniformly random two-dimensional order, whose combinatorics is known [26, 27]; the $e^{-\beta S}$ model of two-dimensional orders also has a well-studied phase structure [26–28]. This paper is built on one exact identity that follows from these facts, and on the use of its violation as a measure of curvature. The *rectangle identity* expresses the spatial distance of two elements through the overlap volumes seen from any element of their common past (Sec. III). It removes the equal-depth requirement of Ref. [10] and yields a distance estimator, which we compare with the equal-depth, past–future and Hu–Meyer–Meyer constructions on the same pairs. In curved spacetime the identity fails as a function of the depth of c; we derive this depth drift in closed form in de Sitter space and to leading order in any $1 + 1$ FRW spacetime, and turn it into a curvature estimator (Sec. IV). Spatial distance in causal sets has also been approached through thickened antichains and induced distances on spatial hypersurfaces [12–14], and curvature through interval and chain abundances in small causal diamonds [6, 15] and the Benincasa–Dowker operator [8]; we compare with the last two at equal density.

As a supporting diagnostic we calibrate the Lorentzian-embedding test of Clough and Evans and add a statistic that lets it reject layered orders on its own (Sec. V), and apply it to sequential growth and to configurations of the $e^{-\beta S}$ model (Sec. VI). Section VII discusses novelty and limitations. Appendix A gives a combinatorial identity for Johnston's propagator at fixed interval size, and Appendix B summarizes other approaches we tested.

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Spatial distances in $3+1$ dimensions, where the rectangle identity has no analogue, are treated in a companion paper [29].

II. SETTING

Test causal sets are generated by sprinkling: a Poisson process of density ρ in a region of a given spacetime, with the order induced by the light cones. The coordinates are then discarded and used only for comparison. We write $\ell = \rho^{-1/2}$ for the discreteness length: a region of volume ℓ^2 contains one element on average.

We use light-cone coordinates $u = t - x$, $v = t + x$, in which $dt\,dx = du\,dv/2$ and $x \prec y$ if and only if $u_x < u_y$ and $v_x < v_y$. The interval between $x \prec y$ is the rectangle $[u_x, u_y] \times [v_x, v_y]$, with volume $V = \Delta u\,\Delta v/2 = \tau^2/2$. A sprinkled causal diamond is therefore a uniformly random two-dimensional order, the intersection of two random linear orders [26, 27]. For two spacelike elements, $d^2 = \Delta x^2 - \Delta t^2 = -\Delta u\,\Delta v$.

The proper time from the longest chain has mean $\simeq 2\sqrt{n}$ in an interval of n elements, with Tracy–Widom fluctuations of order $n^{1/6}$ [30]; wherever a volume is available we use the count instead, $\tau^2 = 2n/\rho$. Pairs of elements in one sprinkling share elements and their errors are correlated; quoted uncertainties are therefore computed from the spread between independent sprinklings.

III. THE RECTANGLE IDENTITY AND A DISTANCE ESTIMATOR

Let a, b be spacelike and c an element in their common past. Following Ref. [10], we split the two intervals into

$$\begin{gathered} A = I(c,a) \setminus I(c,b), \qquad B = I(c,b) \setminus I(c,a), \\ C = I(c,a) \cap I(c,b). \end{gathered} \tag{1}$$

Boguñá and Krioukov choose c at equal proper time below a and b and derive d from the overlap fraction. In $1+1$ dimensions the choice of c is immaterial, because of the following identity (Fig. 1).

Derivation. Place c at the origin and take $u_a < u_b$, $v_a > v_b$ (a and b spacelike). Then $I(c,a) = [0,u_a] \times [0,v_a]$ and $I(c,b) = [0,u_b] \times [0,v_b]$. The rectangle from c to the join (u_b, v_a) splits into four tiles,

$$\begin{aligned} C &= [0,u_a] \times [0,v_b], & A &= [0,u_a] \times [v_b,v_a], \\ B &= [u_a,u_b] \times [0,v_b], & D &= [u_a,u_b] \times [v_b,v_a]. \end{aligned} \tag{2}$$

With $\Delta u = u_b - u_a$ and $\Delta v = v_a - v_b$, the volumes are $V_C = u_a v_b/2$, $V_A = u_a \Delta v/2$, $V_B = \Delta u\, v_b/2$, $V_D = \Delta u\,\Delta v/2$, so that $V_C V_D = V_A V_B$. The tile D has a and b at opposite corners, so $V_D = d^2/2$. Hence, for every c in the common past,

$$d^2 = \frac{2 V_A V_B}{V_C}. \tag{3}$$

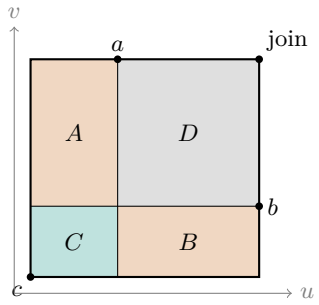


FIG. 1. The rectangle identity in light-cone coordinates (rotated by 45° with respect to the usual picture). The rectangle from c to the join of a and b splits into the overlap C, the strips A and B seen by only one of a, b, and the tile D with a and b at opposite corners. Opposite tiles have equal products, $V_C V_D = V_A V_B$, and $V_D = d^2/2$.

The tile D is the interval between the meet and the join of a and b, i.e., the region whose volume defines the $1+1$ spatial distance of Brightwell and Gregory and of Rideout and Wallden [3, 11]. The identity shows that this volume can be read from the past alone, through a single element c. The equal-time choice of Ref. [10] is $V_A = V_B$, in which case their $1+1$ formula reduces to Eq. (3); for general c the search for equal-time elements, which requires the density and the dimension, is unnecessary. The identity relies on the overlap of two $1+1$ intervals being again an interval; in higher dimensions it is not, and Eq. (3) has no direct analogue.

Estimator. On a causal set, V_X is replaced by N_X/ρ, the number of elements in the region. The three regions are disjoint, so N_A, N_B, N_C are independent Poisson variables with means $\lambda_X = \rho V_X$. The product $N_A N_B$ is unbiased for $\lambda_A \lambda_B$, but $1/N_C$ is biased and undefined at $N_C = 0$. Since $E[1/(N+1)] = (1 - e^{-\lambda})/\lambda$ for a Poisson variable,

$$\widehat{d^2} = \frac{2 N_A N_B}{\rho\,(N_C + 1)} \tag{4}$$

is unbiased for d^2 up to a factor $1 - e^{-\lambda_C}$. The relative variance is approximately $1/\lambda_A + 1/\lambda_B + 1/\lambda_C$, so the deepest available c is optimal; we take the element with the largest N_C. The strips A and B are long and thin and contain many elements even when $d \sim \ell$, which is why the estimator works down to the discreteness scale; by contrast, a direct count of the tile D has relative variance $1/\lambda_D = 2\ell^2/d^2$, which is $\frac{1}{2}$ at $d = 2\ell$.

Tests. Two pair selections are used, and the difference matters for the size of the errors: Table I and Fig. 2 use random spacelike pairs among *late* elements of flat sprinklings of a unit diamond, which have large common pasts; Table II uses pairs in the *middle* of the diamond, which have both a common past and a common future but smaller ones. In both, c is the deepest element for each pair. The median error falls approximately as $N^{-1/2}$ at large d, and remains at 15–25% below 2ℓ, the discrete-

ness scale itself, where $\widehat{d^2}$ is also biased high [by $(37\pm7)\%$ at $N = 4000$ on the pairs of Table II]. Table II compares the estimator with the existing constructions on the same pairs: the equal-depth filter of Boguñá and Krioukov [their Eq. (34)]; the smallest interval $I(c, f)$ with $c \prec a, b \prec f$ (Brightwell–Gregory, Rideout–Wallden), $d^2 = 2(N_{\min} - 2)/\rho$; and the construction of Hu, Meyer and Meyer [19], who showed that for $p \prec a, b \prec q$ with all four proper times equal to s, $d^2 = \tau_{pq}^2 - 4s^2$. They gave it as an exact statement about Minkowski space; on the causal set we read every τ^2 from a count ($\tau^2 = 2N/\rho$), take p from the common past and q from the common future, keep the pairs (p, q) whose four counts agree within Poisson noise ($\chi^2 < 3$), and among them use the one with the smallest counts, which has the smallest variance (choosing instead the best-matched or averaging over all matched pairs is less accurate). Two ablations separate the ingredients of our estimator: the same deepest c without the Poisson correction, $2N_AN_B/\rho N_C$, and the equal-depth elements c of Ref. [10] with the corrected formula, Eq. (4).

Above 2ℓ the rectangle estimator is unbiased for d^2 within the uncertainties. Between 2ℓ and 10ℓ its median error is the smallest; the Hu–Meyer–Meyer estimate is close in error but biased high by about 20%, because $\tau_{pq}^2 - 4s^2$ is a small difference of two noisy terms. Above 10ℓ the Hu–Meyer–Meyer estimate is unbiased and the most accurate, with half the error of ours or less: there the tile between a and b is large, and a direct count of it, corrected exactly by the four equal intervals, beats reading the same volume from the strips. For d itself the rectangle estimator reads 1–3.5% low, as expected from the square root of an unbiased estimate of d^2. The plain past–future count is biased high, because the smallest interval between elements cannot coincide with the tile D. The ablations locate the gain over the equal-depth estimator. With the deepest c, N_C is large and the Poisson correction makes no difference. With the equal-depth elements, where N_C is small, the correction removes the bias of the equal-depth estimator ($+10\% \to +1\%$ and $+6\% \to +2\%$ between 2ℓ and 10ℓ), but not its larger error (0.129 and 0.104, against 0.095 and 0.059 with the deepest c). The bias of Ref. [10] on the causal set thus comes from the uncorrected ratio of small counts, and the accuracy gain from the freedom to use the deepest c, which the identity provides. The equal-depth filter also returns no estimate for 10–30% of the pairs. Below 2ℓ every estimator is poor, and the two past–future constructions fail entirely (median errors above 50%). For the baseline tests we add a bottom element that precedes all others; the deepest c is then always that element, selected without reference to its own count, so the selection introduces no bias (in Table I, without such an element, it is one of the lowest elements); the freedom in the choice of c is exercised by the curvature tests of Sec. IV.

TABLE I. Median relative error of Eq. (4) with the deepest c; flat $1 + 1$ sprinklings, 300 pairs among late elements in each of 4 sprinklings per N. (Table II uses pairs in the middle of the diamond, with smaller common pasts, and therefore larger errors.)

N	$d < 2\ell$	2ℓ–10ℓ	$> 10\ell$
1000	23%	6.5%	5.2%
2000	15%	5.0%	3.4%
4000	19%	5.1%	2.5%

TABLE II. Rectangle estimator, Eq. (4), against the equal-depth filter of Ref. [10], the past–future construction of Hu, Meyer and Meyer [19] read from counts (see text), and the smallest past–future interval [3, 11], on the same spacelike pairs in the middle of flat diamonds (12 sprinklings × 100 pairs per N). Two ablations: the deepest c without the Poisson correction, and the equal-depth elements c with it. Bias: mean relative error of d^2, in percent, with the standard error from the spread between sprinklings in parentheses. Error: median absolute relative error of d. The equal-depth rows have no estimate for 10–30% of the pairs.

		2ℓ–10ℓ		$> 10\ell$	
N	method	bias	error	bias	error
1000	rectangle, Eq. (4)	+1(5)	0.095	−4(4)	0.082
	no correction	+3(5)	0.097	+1(4)	0.080
	equal depth [10]	+10(4)	0.131	+13(3)	0.126
	with correction	+1(3)	0.129	−2(3)	0.127
	Hu–Meyer–Meyer	+23(3)	0.098	+4(2)	0.028
	past–future	+86(3)	0.272	+25(3)	0.095
4000	rectangle, Eq. (4)	0(2)	0.059	−2(1)	0.039
	no correction	0(2)	0.059	0(2)	0.037
	equal depth [10]	+6(2)	0.106	+2(1)	0.075
	with correction	+2(2)	0.104	−5(1)	0.073
	Hu–Meyer–Meyer	+22(4)	0.099	0(1)	0.015
	past–future	+79(5)	0.282	+14(1)	0.049

IV. CURVATURE FROM DEPTH DRIFT

In flat spacetime every c in the common past gives the same estimate. In curved spacetime Eq. (3) no longer holds exactly, and the estimate depends on the depth of c. This dependence measures curvature.

de Sitter volumes. Two-dimensional de Sitter space in conformal coordinates,

$$ds^2 = \frac{-d\eta^2 + dx^2}{(H\eta)^2}, \qquad \eta < 0, \tag{5}$$

has the light cones of flat space. The tiles A, B, C, D are the same rectangles in $(u, v) = (\eta - x, \eta + x)$, but volumes carry the conformal weight,

$$dV = \frac{d\eta\, dx}{(H\eta)^2} = \frac{2\, du\, dv}{H^2(u + v)^2}. \tag{6}$$

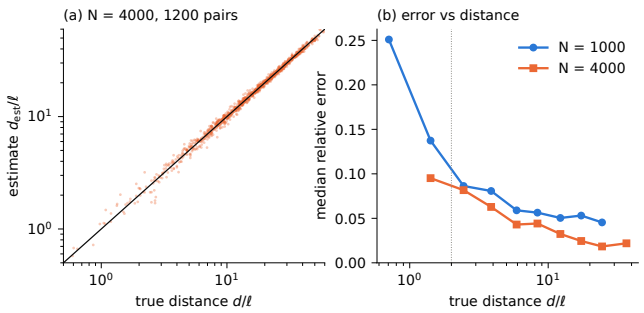


FIG. 2. Eq. (4) with the deepest c on flat $1+1$ sprinklings, from the causal order only. (a) Estimate against true distance, $N = 4000$. (b) Median relative error per distance bin; the dotted line marks 2ℓ.

Integrating over $[u_1, u_2] \times [v_1, v_2]$, the inner integral gives $1/(u+v_1) - 1/(u+v_2)$ and the outer one a logarithm,

$$V_\square = \frac{2}{H^2} \ln \frac{\eta_L\, \eta_R}{\eta_{\rm bot}\, \eta_{\rm top}}, \tag{7}$$

with $\eta = (u+v)/2$ evaluated at the left, right, bottom and top corners (all $\eta < 0$, so the argument is positive).

Closed form. Place a, b on one slice at $(\eta_0, \mp h)$ and c directly below their midpoint at $(\eta_0 - T, 0)$, and set $H = 1$. By symmetry $V_A = V_B$. Writing $\bar\eta = |\eta_0|$ and $q = \bar\eta + T/2$ (so that $|\eta|$ at the corners of A is $q \mp h/2$, $\bar\eta + h$ and $\bar\eta$), Eq. (7) gives, to first order in h,

$$V_A = 2 \ln \frac{(q - \frac{h}{2})(\bar\eta + h)}{(q + \frac{h}{2})\, \bar\eta} \simeq 2h \Big(\frac{1}{\bar\eta} - \frac{1}{q} \Big) = \frac{hT}{\bar\eta\, q}, \tag{8}$$

and at $h = 0$

$$V_C = 2 \ln \frac{q^2}{(\bar\eta + T)\, \bar\eta} = -2 \ln(1-x), \qquad x = \frac{T^2}{4q^2}. \tag{9}$$

The true distance is $d \simeq 2h/\bar\eta$, so

$$\frac{d^2_{\rm est}}{d^2} = \frac{2V_A^2/V_C}{4h^2/\bar\eta^2} = \frac{x}{-\ln(1-x)}. \tag{10}$$

The proper time from c to the midpoint of a, b is $\tau = \ln[(\bar\eta + T)/\bar\eta]$, so that $T/\bar\eta = e^\tau - 1$ and $x = [(e^\tau - 1)/(e^\tau + 1)]^2 = \tanh^2(\tau/2)$. Restoring H,

$$\begin{aligned} \frac{d^2_{\rm est}}{d^2} &= \frac{\tanh^2(H\tau/2)}{2 \ln\cosh(H\tau/2)} = 1 - \frac{(H\tau)^2}{8} + O(\tau^4) \\ &= 1 - \frac{R\,\tau^2}{16} + \ldots, \qquad R = 2H^2, \end{aligned} \tag{11}$$

exact as $d \to 0$ at any depth, for c directly below the midpoint of a and b. The denominator is $H^2/2$ times the de Sitter volume of an interval of height τ, $V = (4/H^2) \ln\cosh(H\tau/2)$, and the result is independent of η_0, as required by the de Sitter symmetry. We verified Eq. (11) against the exact volumes of Eq. (7) to 10^{-5} for $H\tau = 0.1$–3. A sideways offset of c changes it little, and a finite d adds an approximately depth-independent shift. We call $1 - d^2_{\rm est}/d^2$ the depth drift.

Other spacetimes. Every $1+1$ metric is conformally flat, so the leading term should hold locally in any $1+1$ spacetime. We checked this in FRW spacetimes $ds^2 = a(\eta)^2(-d\eta^2 + dx^2)$ with $a = \eta^p$, where $R = 2(\ln a)''/a^2 = -2p\,\eta^{-2-2p}$ varies in time and the rectangle volumes have closed forms. For $p = 1$, -2 and $\frac{1}{2}$ (negative, positive and negative curvature), $16(1 - d^2_{\rm est}/d^2)/\tau^2$ tends to R at the location of a, b as $\tau \to 0$ (to 1% at $\tau = 0.01$), with the correct sign: for $R < 0$ the ratio exceeds 1. At finite depth it measures a curvature between the values at a, b and at c, with more weight near the top. The weighting follows from the next order. Write $a = e^{\sigma(\eta)}$, place a, b at $(\eta_0, \mp h)$ and c a coordinate time T straight below. As $h \to 0$ the strips have $V_A = V_B = h \int_0^T e^{2\sigma(\eta_0 - s/2)} ds$, the overlap is $V_C = \frac{1}{2} \int_0^T \int_0^T e^{2\sigma(\eta_0 - (s+r)/2)} ds\, dr$, and $d = 2h e^{\sigma(\eta_0)}$. Expanding σ in a Taylor series at η_0, integrating term by term, and trading T for the proper time $\tau = \int_0^T e^{\sigma(\eta_0 - s)} ds$ gives

$$1 - \frac{d^2_{\rm est}}{d^2} = \frac{\tau^2}{16} \Big[R - \frac{\tau}{3} \frac{dR}{d\tau} + O(\tau^2) \Big], \tag{12}$$

with $R = 2\sigma'' e^{-2\sigma}$ and its derivative along the worldline (toward the future) evaluated at a, b. To this order the drift measures R one third of the way from a, b down to c, which is the mean depth of a triangular weight $2(\tau - s)/\tau^2$ on the depths $s \in [0, \tau]$. The exact power-law volumes confirm Eq. (12): for $p = 1$ at $\eta_0 = 1.5$, $16(1 - d^2_{\rm est}/d^2)/\tau^2 = -0.3982$, -0.4073, -0.4200 at $\tau = 0.02$, 0.05, 0.1, against -0.3997, -0.4068, -0.4185 from its two terms ($R = -0.395$ at a, b); the other two values of p agree equally well.

Test on sprinklings. We sprinkle de Sitter space ($H = 1$, $\rho = 10^4$) and, as a control, flat space in the same coordinates, insert a, b by hand $\approx 14\ell$ apart, and evaluate Eq. (4) for every sprinkled c in their common past whose rectangle lies inside the region. Figure 3 shows the mean ratio per depth bin over 400 sprinklings against the continuum value computed for the same elements c, which includes the effects of finite d and of sideways offsets and therefore lies slightly below Eq. (11). The flat control remains at 1 and de Sitter follows the continuum. Neighboring depth bins share elements: computed from the spread over the 400 sprinklings, the correlation of adjacent bins is 0.80, 0.81, 0.90, 0.94, 0.96 in de Sitter space (from the shallowest pair to the deepest) and 0.80–0.97 in flat space, and bins two apart are still correlated at 0.5–0.9. A run of deviations in adjacent bins therefore counts as a single fluctuation, and the points of Fig. 3 must not be combined as independent. In the shallowest bin the tile C contains few elements, the factor $1 - e^{-\lambda_C}$ of Eq. (4) is not negligible, and the estimator reads low in both geometries.

A curvature estimator. For one pair a, b we fit $\widehat{d^2} = \alpha + \beta\tau^2$ over the elements c of their common past, with

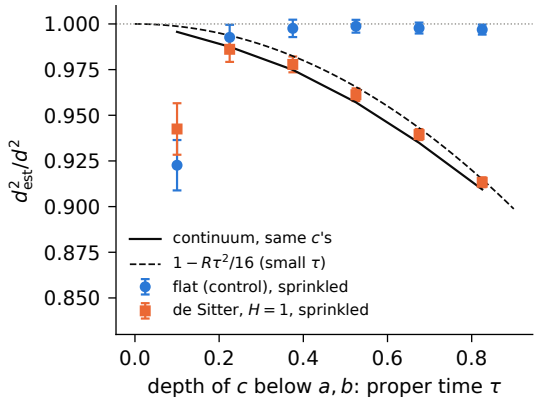


FIG. 3. Depth drift of the rectangle estimate: mean over 400 sprinklings ($\rho = 10^4$, $H = 1$) per bin of the proper time τ from c to a, b, with errors from the spread between sprinklings. The bins are strongly correlated (adjacent bins 0.8–0.97; see text), so the error bars are not independent. Solid: continuum ratio from the exact volumes, averaged over the same elements c (finite d and offsets included). Dashed: small-τ term of Eq. (11).

$\tau^2 = (N_{I(c,a)} + N_{I(c,b)})/\rho$ from counts, and read $R = -16\beta/\alpha$. Two choices need care.

Window from counts. The fit uses every c with $N_C \geq 20$, which keeps the factor $1 - e^{-\lambda_C}$ of Eq. (4) negligible, down to the deepest c whose rectangle lies in the sprinkled region. Neither limit refers to H. Because this window reaches $H\tau \approx 1$, the τ^4 term of Eq. (11) contributes, and the same fit applied to the exact continuum volumes of the same elements gives $R^{\rm cont}_{\rm fit} = 2.07$–$2.08$ in de Sitter space instead of 2; this, not 2, is the value the estimator should reproduce. Varying the lower cut ($N_C \geq 5$, 20, 50) and the upper cut (τ^2 up to 50%, 75% or 100% of its largest value) changes the pooled de Sitter result only between 1.91 and 2.01 at $\rho = 10^4$ (flat: -0.01 to 0.00), within 0.06–0.28 of the continuum value of each rule; the largest offset belongs to the shortest window, which is also the noisiest.

Pooling. A single pair gives a noisy estimate (standard deviation 3.5 at $\rho = 4000$ and 2.4 at $\rho = 10^4$ in de Sitter space), and the mean of single-pair values is biased low: the intercept and slope of one least-squares fit have anticorrelated errors, so $E[-16\beta/\alpha]$ is smaller than $-16E[\beta]/E[\alpha]$. This ratio bias is the negative offset that version 1 of this paper could not exclude. Averaging the fit coefficients over the pairs before forming the ratio, $R_{\rm est} = -16\langle\beta\rangle/\langle\alpha\rangle$, removes it to first order. Table III shows the pooled estimator over 1000 sprinklings with one pair each, with bootstrap errors. With inserted pairs it agrees with the continuum value of the same fit, in flat space within the errors and in de Sitter space within 1.2σ at both densities. With sprinkled pairs, where a and b are the sprinkled elements nearest to the two target points, it reads low in de Sitter space, by 0.37 ± 0.11 at $\rho = 4000$ and 0.24 ± 0.07 at $\rho = 10^4$; the shortfall decreases with density, but we have not identified its cause. In the $p = 1$ FRW spacetime ($\rho = 10^4$, 200 sprinklings, a, b at $\eta_0 = 1.5$, the window of version 1), the binned ratio follows the exact continuum within 1σ for $\tau \geq 0.15$, and the mean of single-pair fits (version 1, before pooling) gives -1.25 ± 0.30 against -0.81 for the continuum fit, while R ranges from -0.4 at a, b to -2.8 at the bottom of the window: the estimator measures a depth-weighted curvature, as Eq. (12) says.

TABLE III. Pooled curvature estimator $R_{\rm est} = -16\langle\beta\rangle/\langle\alpha\rangle$ over 1000 sprinklings with one pair each ($H = 1$; count-defined window). Errors: bootstrap over pairs. $R^{\rm cont}_{\rm fit}$: the same fit on the exact continuum volumes of the same elements. Last column: standard deviation of single-pair values (de Sitter).

ρ	pairs	flat	de Sitter	$R^{\rm cont}_{\rm fit}$	single pair
4000	inserted	-0.03 ± 0.02	1.97 ± 0.11	2.07	3.5
	sprinkled	-0.07 ± 0.02	1.70 ± 0.11	2.07	3.6
10^4	inserted	-0.01 ± 0.02	1.99 ± 0.07	2.08	2.4
	sprinkled	-0.04 ± 0.02	1.84 ± 0.07	2.08	2.4

Comparison with other estimators at equal density. On the same de Sitter and flat sprinklings we evaluated two established routes. The Benincasa–Dowker operator [8], in its nonlocal two-dimensional form with smearing parameter ϵ, applied to a constant field at the position of the pair's midpoint, gives $R = -2B_\epsilon[1]$: at $\rho = 4000$ its single-point standard deviation is $\approx 1.2 \times 10^3$, 160 and 23–30 for $\epsilon = 0.1$, 0.03 and 0.01, and at $\rho = 10^4$ it is larger (72–82 at $\epsilon = 0.01$), as expected at fixed ϵ. Stronger smearing lowers the spread (2.7–4.5 at $\epsilon = 0.003$), but then the operator needs the past light cone far beyond our sprinkled region, and the truncation dominates its mean (+0.7 in flat space, +4.7 in de Sitter space); a fair comparison at small ϵ needs a much larger region, which we have not run. The ordering fraction of a causal diamond, whose curvature correction [6] we calibrate on the exact de Sitter continuum, gives for one diamond of height 0.7 a standard deviation of 3.2 at $\rho = 4000$ (about 10^3 elements) and 2.1 at $\rho = 10^4$, unbiased in both geometries (2.06 ± 0.22 and 1.87 ± 0.15 in de Sitter, -0.10 ± 0.23 and -0.05 ± 0.13 in flat space). The depth drift of one pair, which uses a common past of comparable size, has a standard deviation of 3.5 and 2.4 at the same densities. Our estimator is therefore not less noisy than the chain-abundance route; it is much less noisy than the local or weakly smeared Benincasa–Dowker operator, whose purpose is the action rather than a pointwise curvature estimate. Its distinctive features are geometric: the curvature appears as the failure of an exact flat-space identity, and Eq. (12) says where in the region it is measured. Other counting routes use Ollivier–Ricci curvature [31], which we have not compared.

V. A SUPPORTING DIAGNOSTIC: THE EMBEDDING TEST AND ITS BLIND SPOT

Embedding. Given the squared intervals s_{ij} between M sampled elements, classical multidimensional scaling (MDS) with a Lorentzian Gram matrix reconstructs coordinates. Clough and Evans applied it to causal sets and proposed its eigenvalue spectrum as a test of dimension and manifoldlikeness [20, 21]. With $s_{ij} = (x_i - x_j)^\top \eta\, (x_i - x_j)$, double centering, $G = -\frac{1}{2} JSJ$ with $J = I - \mathbf{1}\mathbf{1}^\top/M$, gives $G = X\eta X^\top$ for the centered coordinates X. A $1+1$ Lorentzian set shows one large negative eigenvalue (time), one large positive eigenvalue (space), and the others near zero; the two eigenvectors give coordinates up to a Lorentz transformation.

We supply MDS with counts only: $s = -2N(I)/\rho$ for related pairs and Eq. (4) with the deepest c for spacelike pairs. On $M = 150$ elements of a flat sprinkling with $N = 2000$, positions are recovered to $0.43 \pm 0.07\,\ell$ across a region $\approx 38\ell$ wide, and the best affine map from the reconstructed to the true coordinates, $x_{\text{true}} \approx A x_{\text{rec}} + b$, is a Lorentz transformation to within 3–10%, in the sense that the largest entry of $A^\top \eta A - \eta$ is 0.03–0.10 in absolute value. The two large eigenvalues are $\approx \mp 2.2$ and all others are below 0.1, in the coordinate units of the sprinkled region (a unit square in u, v).

Statistics and calibration. As test statistics we use intrinsic numbers only: r_+, the ratio of the second to the first positive eigenvalue; the share of the spectrum in one time and one space direction, $(|\lambda_-| + \lambda_+)/\sum_i |\lambda_i|$; and the *stress*, the rms misfit of the squared intervals by the best $1+1$ Minkowski embedding relative to their rms. The sample consists of the $M = 40$ elements with the largest pasts, after removing elements with no common past with the rest. Figure 4(a) shows the calibration on flat universes (random two-dimensional orders): stress and r_+ settle only for $N \gtrsim 100$ (stress 0.16, 0.10, 0.08 at $N = 100$, 200, 400), so the test is used only above this size. Curvature raises the stress: for de Sitter diamonds with fixed $N = 400$ it stays in the flat band up to $H\tau \approx 1$ and reaches 0.25 at $H\tau = 4$ [Fig. 4(c)]. The stress thus tests whether the order is a *flat* sheet, while Sec. IV measures its curvature.

Blind spot. Applied to Monte Carlo samples of the $e^{-\beta S}$ model at $N = 400$ (Sec. VI), this test passes every continuum-like state and fails orders with 4 and 7 layers, but it does not reliably flag orders with about 10 layers: at $M = 40$ one of seven lies in the flat band and two more just above it, and at $M = 80$ two others have stress 0.03, lower than flat orders. Two effects are responsible. The sample contains only the top one or two layers. And in a layered order most sampled pairs have zero squared interval: linked pairs have $N(I) = 0$, and spacelike pairs often have an empty strip, $N_A = 0$ or $N_B = 0$. MDS fits an all-zero matrix perfectly; a bilayer produces exactly that.

Null-pair fraction. We add the fraction z of sampled pairs with zero squared interval, taken as the larger of its values on the top sample and on a sample spread evenly in height (longest chain below the element) over the upper 70% of the order. On sprinklings z decreases with N (median 0.18, 0.12, 0.09 at $N = 100$, 200, 400) and does not respond to curvature: de Sitter diamonds at $H\tau = 4$ give 0.08–0.11. Layered orders give 0.4–1 [Fig. 4(b)]. We flag an order as layered when z exceeds 1.5 times the flat median at the same N. Orders with only 2–3 layers cannot be processed (too few elements share a common past), and the rule counts them as flagged. *What z adds.* Layered orders are also detected by simpler statistics: the height (number of layers), the ordering fraction and the Myrheim–Meyer dimension [4, 5], interval abundances [7], and graph observables [39]; in the $e^{-\beta S}$ literature the phases are characterized by the action, the ordering fraction, the height and interval abundances [16–18, 27, 28]. On our states the fixed-size ordering fraction separates the same groups as z (Appendix A). The null-pair fraction does not detect anything these miss. Its role is to make the embedding test usable on its own: without it the test passes some 10-layer orders, so a user who relied on the embedding spectrum alone would accept them. With z the test needs no second statistic, and it uses only quantities the embedding already computes.

VI. APPLICATIONS

Sequential growth. In transitive percolation, the simplest classical sequential growth model [40], elements are born one at a time and each newborn element chooses each existing element as an ancestor with probability p, and the order is the transitive closure. As a sanity check of the diagnostics we grew universes of $N = 100$ and 200 elements with $p = 0.05$, and with the non-covariant variant $p_n = 3/n$. Neither produces a $1+1$ spacetime: the ordering-fraction dimension drifts with scale, and the embedding test fails on every statistic (share of the spectrum in one time and one space direction ≈ 0.33 against 0.65 for flat orders at $N = 100$; stress 0.64–0.78 against 0.10–0.12 at $N = 200$). Their null-pair fraction is not what flags them; grown universes with $N = 1500$ also fail the propagator target (Appendix A). Covariant approaches to cosmology in growth models are discussed in Refs. [41, 42].

The two phases of the $e^{-\beta S}$ model. Weighting two-dimensional orders by $e^{-\beta S}$, with S the Benincasa–Dowker action, gives a first-order transition between a continuum-like and a layered phase [26–28]. We sampled it with Metropolis swap moves at nonlocality parameter $\epsilon = 0.21$ (an $O(N^2)$ update, since a swap changes two rows and columns of the causal matrix). Table IV shows the diagnostics at $N = 200$. The states at $\beta \geq 0.6$ are metastable intermediates; the ground state at large β is a bilayer [43].

The states below are test configurations, representative of the two phases, not equilibrium samples (see the end of this section). We applied the flag to 44 final states

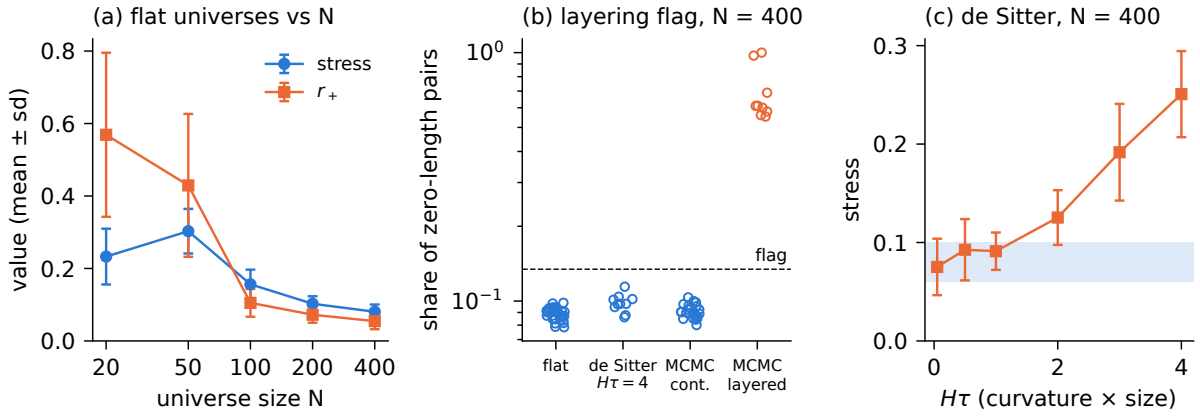


FIG. 4. Calibration of the embedding test. (a) Flat universes (random two-dimensional orders, 10 per size, $M = 40$): stress and r_+ against the number of elements N. (b) Null-pair fraction z at $N = 400$ ($M = 80$) for flat orders, de Sitter diamonds at $H\tau = 4$, and final states of $e^{-\beta S}$ Monte Carlo chains (continuum-like, 30–39 layers; layered, 4–11 layers). Dashed: the flag threshold, 1.5 times the flat median. The 13 states with 2–3 layers are not shown. (c) Stress of de Sitter diamonds at fixed $N = 400$ (8 per height) against $H\tau$; shaded: the flat band (±1 standard deviation).

TABLE IV. Embedding diagnostics on $e^{-\beta S}$ states at $N = 200$ ($\epsilon = 0.21$). f: ordering fraction; L: number of layers. The last column gives the range of the null-pair fraction z on fresh chains, for $\beta \le 0.3$ (second row) and $\beta \ge 0.6$ with 8–13 layers (third row).

$N = 200$	f	L	r_+	stress	z
random orders	0.50	24	0.07	0.10	0.10–0.15
$\beta = 0.05$	0.51	24	0.10	0.11	0.12–0.13
$\beta = 0.6$	0.59	9	0.39	0.28	0.39–0.48

of $N = 400$ chains run from random and from layered initial orders at $0.01 \le \beta \le 0.7$. All 22 continuum-like states (30–39 layers) pass; all 9 states with 4–11 layers are flagged ($z = 0.55$–1.0 against a threshold of 0.13); the 13 states with 2–3 layers cannot be processed and are counted as flagged. The fixed-n ordering fraction (Appendix A) separates the same groups; the propagator statistic misses one 4-layer state. Because the flag was designed after these states had been seen, we tested it on held-out samples. On fresh $N = 200$ chains, 12 of 12 continuum-like states pass and 18 of 18 layered states ($\beta = 0.6$–1.5) are flagged. At $N = 800$, with the threshold fixed in advance from flat orders ($z > 0.120$; none of 50 fresh flat orders and 20 de Sitter diamonds is flagged), [r27: results of 28 chains, $\beta \in \{0.01, \ldots, 0.7\} \times$ 4 seeds, random initial orders, 2000 sweeps]. These states are test inputs, not equilibrium samples: chains started from layered orders accept $\le 0.1\%$ of the moves. They bracket the transition only loosely: chains from layered starts become continuum-like at $\beta \le 0.05$, and chains from random starts become layered at $\beta \ge 0.45$. The bracket $0.05 < \beta_c(400) < 0.45$ is compatible with the finite-size scaling $\beta_c \approx 1.66/(\epsilon^2 N) = 0.094$ of Ref. [28]; it is too loose to test it.

VII. DISCUSSION

What is new. The overlap regions are those of Ref. [10], the embedding test is that of Ref. [20], the null-pair fraction is related to Ref. [7], and the Laguerre identity is classical combinatorics. To our knowledge the following are new: the rectangle identity for general c, Eq. (3), with the bias-corrected estimator and its comparison with existing constructions, including a counting version of the Hu–Meyer–Meyer construction; the depth drift, its closed form in de Sitter space, Eq. (11), and its leading form in FRW spacetimes, Eq. (12), with the curvature estimator built on it; the application of the Laguerre identity to Johnston's propagator, Eq. (A2); and the calibration of the embedding test against size, curvature and layering. Our literature search (October 2026) covered the citing papers of Refs. [10, 20, 24, 25]; the rectangle identity is elementary enough that it may appear in passing in work we did not reach.

Limitations. The exact results are specific to $1 + 1$ dimensions; the rectangle identity has no direct analogue in higher dimensions, where the overlap of two intervals is not an interval. The distance estimator is the most accurate only below about 10ℓ; at larger separations the counting version of the Hu–Meyer–Meyer construction is better. The curvature estimator must be pooled over many pairs (single pairs are dominated by noise and their ratio is biased), was tested on de Sitter and one FRW spacetime, measures R at about a third of the depth of c, and with sprinkled rather than inserted pairs reads low by 0.2–0.4 in de Sitter space at the densities used. The embedding thresholds are empirical and must be recalibrated at each N; the $e^{-\beta S}$ states used are test inputs,

not equilibrium samples. The Laguerre target is exact only for two-dimensional orders, and as a manifoldlikeness diagnostic it is not stronger than the ordering fraction.

Outlook. Since the Hu–Meyer–Meyer construction is the more accurate one at large separations, a depth drift built on it may be less noisy there; a combined distance estimator would switch between the two near 10ℓ. A fixed-n propagator target in $3+1$ dimensions would not be exact: Johnston's $3+1$ particle hops along links only [24], so a path requires every skipped sub-interval to be empty, and $E[K\,|\,n]$ becomes a sum of $(k-1)$-point integrals with factors $(1-W/V)^{n-k+1}$. Its first term can be evaluated at large n: the expected number of two-hop link paths tends to 3π, which is what Johnston's amplitudes require to reproduce the short-distance limit $m^2/8\pi$ of the continuum mass term [24], so this is a consistency check rather than a new result; a Monte Carlo evaluation approaches it as $3\pi(1-1.6/\sqrt{n})$ at fixed n. A $3+1$ target would therefore be a computed table rather than a formula.

DATA AVAILABILITY

The code that produces every number and figure in this paper, and the research notebook in which the work was carried out, are openly available at `https://github.com/presindev/causet-counting`; the version used for this paper will be archived with a persistent identifier [Zenodo DOI] on submission. Figures are produced by `sims/fig_part1.py`; the baseline comparison by `sims/r31_rect_baseline.py` and, with the Hu–Meyer–Meyer construction and the ablations, `sims/r36_rect_ablation_hmm.py`; the curvature estimator (count-defined window, pooled form, sprinkled pairs, Benincasa–Dowker and chain-abundance comparison, bin correlations) by `sims/r37_curvature_referee.py`; Eq. (12) by `sims/r38_drift_kernel.py`; the null-pair fraction by `sims/r8_detector_spread.py`; the curvature fit and the FRW check by `sims/r9_drift_fit_frw.py`; the $e^{-\beta S}$ states by `sims/n400_crystal.py`; the propagator statistics by `sims/b_johnston_n400.py` and `sims/r32_laguerre_vs_fraction.py`; the $N = 800$ test by `sims/r27_n800_detector.py`.

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Appendix A: A density-free target for Johnston's propagator

This appendix records a combinatorial identity for Johnston's propagator. As a diagnostic of manifoldlikeness it is not stronger than the ordering fraction (end of this appendix); its use is as an exact, density-free check of implementations of the chain sum.

The propagator. In Johnston's model a particle hops along the order with amplitude a per hop and b per intermediate stop [24]. With C the causal matrix ($C_{xy}=1$ if $x \prec y$) and $\Phi = aC$,

$$K = \Phi + b\,\Phi^2 + b^2\Phi^3 + \cdots = \Phi\,(I - b\Phi)^{-1}, \tag{A1}$$

with $a = \frac{1}{2}$ and $b = -m^2/\rho$ in $1+1$ dimensions. $K(x,y)$ is a weighted count of chains, $K(x,y) = \frac{1}{2}\sum_k c_k(x,y)\,(-\mu/2)^k$, where c_k is the number of k-element chains strictly inside $I(x,y)$ and $\mu = m^2/\rho$. In a sprinkled interval of volume $V = \tau^2/2$ the expected number of k-chains is $(\rho V)^k/(k!)^2$, so that $\langle K\rangle = \frac{1}{2}\sum_k[-(m\tau/2)^2]^k/(k!)^2 = \frac{1}{2}J_0(m\tau)$, the retarded Klein–Gordon propagator [24].

Fixed interval size. K depends only on the order and on μ. In a uniformly random two-dimensional order the n elements inside an interval again form a uniformly random two-dimensional order, and any k of them form a chain with probability $1/k!$ (the two linear orders must agree on them). Hence $E[c_k\,|\,n] = \binom{n}{k}/k!$ [32], and

$$\begin{aligned} E[K\,|\,n] &= \frac{1}{2}\sum_{k=0}^{n}\binom{n}{k}\frac{(-\mu/2)^k}{k!} = \frac{1}{2}L_n\!\left(\frac{\mu}{2}\right) \\ &\approx \frac{1}{2}e^{\mu/4}J_0\!\left(\sqrt{2\mu(n+\tfrac{1}{2})}\right), \end{aligned} \tag{A2}$$

a Laguerre polynomial in the interval size; the Bessel form is the Hilb-type asymptotic for large n at fixed μ [33]. Averaging over Poisson-distributed n with mean $\lambda = \rho\tau^2/2$, using $\sum_n e^{-\lambda}\frac{\lambda^n}{n!}\binom{n}{k} = \frac{\lambda^k}{k!}$, recovers $\frac{1}{2}\sum_k(-\mu\lambda/2)^k/(k!)^2 = \frac{1}{2}J_0(m\tau)$: Johnston's result is the Poisson average of Eq. (A2). The fixed-n form requires neither the density nor coordinates: every related pair of a causal set supplies its n and its K, and Eq. (A2) is the target. The identity itself is combinatorial: at $\mu = -2$ and summed over permutations it counts all increasing subsequences, $n!\,L_n(-1)$ [34, 35]. A fixed-n test also exists for maximal chains, whose length distribution in a two-dimensional interval of n elements measures dimension and embeddability [36]; Eq. (A2) instead sums all chains with the propagator's weights. Path-sum studies of the propagator average over sprinklings [37, 38].

Tests. Figure 5 bins all related pairs of an $N = 400$ order by n and compares $\langle K\rangle$ with Eq. (A2) at $\mu = 0.05$. As a statistic we use the rms deviation S_K of K from the target over all related pairs with $n \geq 2$, and compare it with its distribution over 40 random orders ($S_K = 0.0158 \pm 0.0018$). We apply it to 60 final states of $N = 400$ chains of the $e^{-\beta S}$ model (Sec. VI; the 44

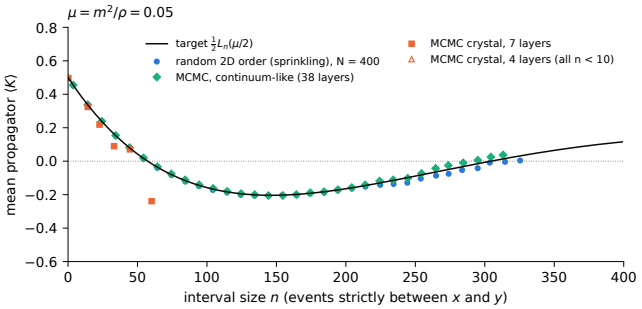


FIG. 5. Johnston's propagator averaged over related pairs with interval size n, against $\frac{1}{2}L_n(\mu/2)$, at $\mu = 0.05$ and $N = 400$. At large n few pairs remain and the bins are noisy. In the 4-layer crystal every interval has fewer than 10 elements, so it contributes a single point near $n = 0$.

states used there and 16 from chains completed later): 28 continuum-like states (30–39 layers), 14 states with 4–11 layers, and 18 with 2–3 layers, which contain too few large intervals for the statistic. Continuum-like states lie within 3 standard deviations of the random orders. States with 9–11 layers deviate by 39–62 standard deviations. In 4- to 7-layer crystals no interval contains more than a few elements, so only small n, where the target is close to $\frac{1}{2}$, is probed: their deviations range from -9 to $+25$ standard deviations, and one 4-layer state lies within 1.2 standard deviations of the random orders. S_K alone therefore does not flag every crystal. Sequentially grown universes ($N = 1500$) deviate strongly, one ($p = 0.02$) reaching $|K| \approx 10$ for individual pairs, whereas the target is bounded by $\frac{1}{2}$; since such orders are not two-dimensional orders, this only confirms that the target is not met by arbitrary orders.

Relation to the ordering fraction. Expanding Eq. (A2) in μ, the first nontrivial term is c_2, the number of related pairs inside the interval, whose fixed-n mean $\binom{n}{2}/2$ is the ordering fraction behind the Myrheim–Meyer dimension [4, 5]. We therefore compared S_K with the analogous statistic S_f, the rms deviation of $c_2/\binom{n}{2}$ from $\frac{1}{2}$, on the same states. The layered states deviate from random orders by 50–58 standard deviations in S_f, comparable to S_K, and all six 4- to 7-layer crystals by more than 100, including the one that S_K misses. As a diagnostic of manifoldlikeness, the propagator target is therefore not stronger than its first term. Its use lies elsewhere: it is an exact, density-free statement about the propagator itself, against which an implementation of the chain sum can be checked pair by pair on any two-dimensional order, without knowledge of the density.

Appendix B: Other approaches tested

- *Fixed-n propagator as a curvature estimator.* In curved $1+1$ spacetime a term $(R\tau^2/48)J_2(m\tau)$ appears in the propagator [44]. In a de Sitter diamond conditioned on n elements the order depends only on $(n, H\tau)$, so a fixed-n curvature test exists. We find that the fraction of related pairs rises by $\approx 0.027\,(H\tau)^2$ for $n = 10$–60, and that the fixed-n propagator is no more sensitive than that fraction alone: a 3σ detection at $n = 30$, $H\tau = 1$ requires 175 intervals in either case. The fixed-n propagator therefore detects curvature only through the known chain-abundance route [6].

- *Embedding test on small orders.* We first applied the test to exhaustive ensembles of small orders ($N \leq 12$). Its statistics do not separate flat from layered orders at these sizes, and only settle for $N \gtrsim 100$ [Fig. 4(a)], so we moved to Monte Carlo samples at $N \geq 200$.

- *Hysteresis in the $e^{-\beta S}$ chains.* At $N = 400$, chains started from random and from layered orders at the same β end in different phases. We first read this as hysteresis of the transition; the chains from layered starts accept $\leq 0.1\%$ of the moves and are effectively frozen, so the difference reflects the sampler rather than the model.

- *Two-dimensional estimator on $2+1$ sprinklings.* Fed into the embedding, Eq. (4) applied to a $2+1$ sprinkling (with the $2+1$ density) produces two large positive eigenvalues. The formula is not valid there, so this is at most a qualitative indication that the spectrum counts spatial dimensions.

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